

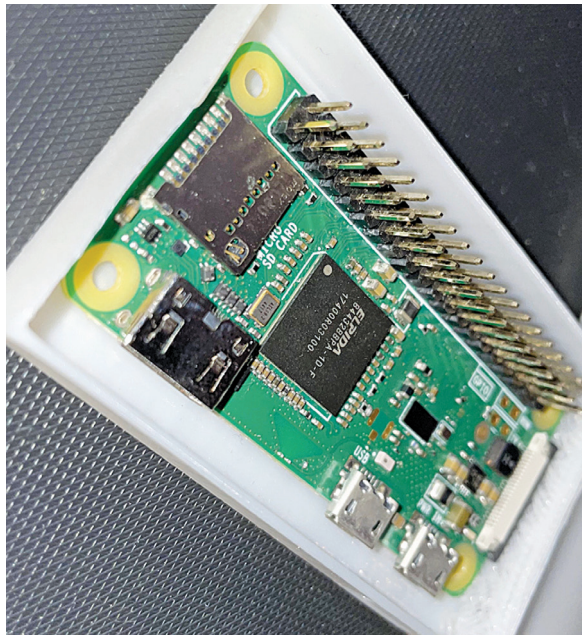


Fig. 14: E-ink header-to-header pins on Raspberry Pi

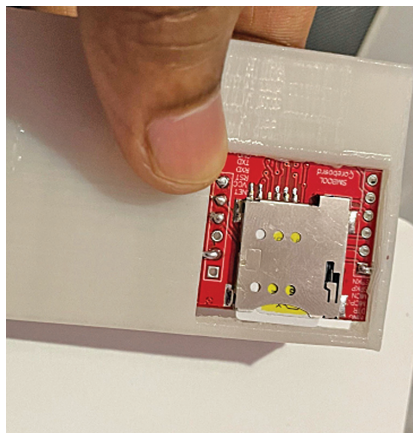


Fig. 15: GSM module fixed in the enclosure



Fig. 16: Phone screen fixed on the base phone cover

mand to GSM module to call that number. It also sets the page number to the calling screen and updates the display with calling screen.

In the calling screen, check the touch points for Call Hang icon. If your touch point matches with the range of call-hang icon, touch point will send the command to SIM800L GSM module to cut the call and hang up. It will also change the page number to 0 and update the display with home screen UI.

Similarly, check the touch points and incoming call and then change the page number and update the display screen with UI for the message, contact, or ringing screen. Then in that screen, check the touch point for the button icon, touch point range, and set the function for those buttons—like hanging phone, home, main screen, calling, and other functions for the button icons. The Python code for achieving the above-mentioned actions are shown in Fig. 9 through Fig. 12.

Connection

After coding, connect the components as per the diagram shown in Fig. 13. Place the Raspberry Pi and SIM 800L module inside in the enclosure/body that has been designed. Fix the speaker and microphone against the holes made in the enclosure for them.

Fix the battery and charging unit in between the display and Raspberry Pi. Then mount the display on the enclosure with the pins matching the E-ink female header-to-header on Raspberry Pi, as shown in Fig. 14. Fix the GSM module on the slot, as shown in Fig. 15, and the phone

EFY Note

The source code of this project is available for free download at www.electronicsforu.com

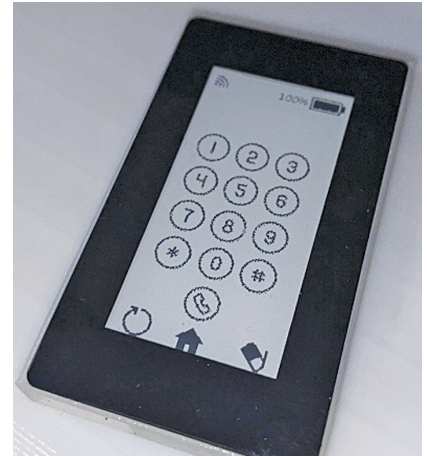


Fig. 17: E-ink phone after assembly

screen on the base phone cover, as shown in Fig. 16.

Testing

The E-ink phone is now ready to show its magic. Insert the SIM in GSM module and run the OS code, and you are ready to use it.

Tap on on the Call icon and you get the dialing screen. Key in a phone number and touch on Call icon to make the call. Touch the Home icon to get the home screen and touch Contacts to see the contact numbers.

Next, try making a call from some other phone to your E-ink phone. When connected, the buzzer starts ringing and it takes you to the ringing screen. Here, touch the Call Attend icon to attend the call and Call Hang icon to hang up the call.

The final assembled E-ink phone is shown in Fig. 17. Enjoy! **EFY**

Note. This is the first version of the phone. The author keeps adding new features and improvements on www.electronicsforu.com

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